Architectural Particular Specification

GYPSUM BLOCK WALL

SECTION 3.2

Rev	Date	Description
0	14 APR 2022	First Issue

Proposed Data Centre Development
at 2-10 Tai Yuen Street,
Kwai Chung, N.T.
Andrew Lee King Fun & Associates Architects Limited

SECTION 3.2

GYPSUM BLOCKWALL

PART 1 GENERAL

1.01 SUMMARY OF WORKS

The works shall include supply and installation of AUGREEN Gypsum Block Wall system, or approved equivalent, including water repellent gypsum block wall, stiffening steel post and beam, dowel bar, gypsum based adhesive, gypsum filler, fire resistant sealant and the gypsum block work accessories such as ties, anchors and isolation joints with adjacent structural elements.

1.02 REFERENCE

(NOT USED)

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. Works shall be detailed and installed to achieve adequate and satisfactory performance in accordance with BS 5628 / PD 6697, BS EN 1996-3, BS EN 1996-2, BS EN 1996-1-1 (Code of practice for the use of masonry), BS 8000 (Workmanship on building sites) and BS 6399 / BS EN 1991-1-1, BS EN 1991-1-7 (Loading for buildings) with regard to building use category.

   The manufacturer of the proposed Gypsum Block Wall system shall be approved by the Architect.

B. Technical

   1. The standard gypsum block shall have a minimum average compressive strength of 5MPa.
   2. The gypsum blocks shall meet fire rating of 2 hours and 4 hours (for 75mm / 80mm and 100mm / 150mm thick respectively) in accordance with BS476 Part 22.
   3. Water absorption rate of water repelling gypsum blocks shall not exceed 0.5% (H1 Class).
   4. The gypsum block wall system shall comply with the requirements of BS5234 part 2 1992.

5. For achieving specific STC rating, the works shall follow the installation models of AUGREEN Eco Quiet System which were tested in accordance ASTM E-90-04 “Laboratory Measurement of Airborne Sound Transmission Loss of Building Partitions”.

Model	STC	Total Thickness (mm)
AUGREEN EQ37	37	80
AUGREEN EQ38	38	75
AUGREEN EQ40	40	75
AUGREEN EQ41	41	100

6. The AUGREEN Block wall system shall comply with the requirements of BS5234 part 2 1992

   a. Determination of Partition Stiffness-
   Maximum deflection of 0.33mm with Apply Load of 500N

   b. Surface Damage of a Wall Partition by Small Hard Body Impact
   Depth of Indentation not exceed 4.99mm with an impact energy of 6Nm.

   c. Resistance of a Wall Partition to Damage by Impact from Large Soft Body
   Permanent Deformation of not exceeding 0.13mm

   d. Resistance of a Wall Partition to Structural Damage by Multi-Impacts
   No collapse of Dislocation with Impact Energy of 120Nm.

   e. Resistance to Crowd Pressure
   No Collapse or Damage with Applied Load of 1.875KN or (0.75kN/m)

   f. Light Weight Anchorage Pull Out Test
   No Release of Shim Plate or Damage to the Partition

   g. Heavy Weight Anchorage (Wash Basin)
   Deformation not exceed 0.56mm observed at a loading

   h. Heavy Weight Anchorage (High Level Wall Cupboard)
   Test Grade of Heavy Duty (4,000N)

   i. Effects of Door Slamming
   Test Grade of Medium Duty

7. The content of calcium sulfate dihydrate in the high density gypsum block shall not be lower than 95% by weight as tested in accordance with BS 1191-1:1973.

8. The content of chloride in the high density gypsum block shall be 0.1% by weight or less as tested in accordance with CSI:2010.

9. The pH value of the high density gypsum block shall be within pH 5 to 9 as tested in accordance with Geospec 3:2001, C1.9.5 at the temperature of 22°C.

1.05 ENVIRONMENTAL REQUIREMENTS

A. The gypsum block shall be an environmental product declaration following ISO 14025 and the drafts of CEN/TC 59 Building Construction SC 17 being the Sustainability in building construction.

B. The gypsum block shall be fulfill the emission test in compliance with DIN EN ISO 16000-9/-11.

C. The gypsum block shall be tested and complied the life cycle assessment in accordance to DIN ISO 14040.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. Calculations
Calculations shall be prepared by the Contractor's Engineer and verified by the Contractor's Register Structural Engineer confirming that the Gypsum Block Wall systems have been designed to satisfy the specified performance requirements, in particular regard to the specified loading criteria and maximum deflection limits.

B. Shop drawing
The Contractor shall be responsible for the preparation of shop drawing with all demarcation plans, installation details and interfacing details to the Architect for approval before placing orders.

C. Method Statement
Before placing orders with suppliers, the Contractor shall submit to the Architect for approval, method statement to fully describe the tools and materials to be employed; method, timing and sequences of mixing, installation, curing, inspection of the gypsum block wall system; all in accordance with the specifications, manufactures' recommendation and safety recommendations.

D. Steel stiffeners for gypsum block walls
Provide steel stiffeners as designed by the Contractor's Engineer. Relevant calculations to be submitted for the approval by Architect / Engineer.

E. Supplier's Certificate of Regional Manufacturing
The Contractor shall submit a supplier's certificate with relevant supporting document to the Architect to prove that the GYPSUM Block Wall system is manufactured locally within 800km from the site.

F. Certificate of Recycle Raw Material
The Contractor shall submit a supplier's certificate with relevant supporting document to the Architect to prove that the gypsum block wall system contain at least 50% in weight of recycled materials, including but not limited to chemical gypsum generated by Flue Gas Desulphurization (FGD) Plant of Electrical Power Plant.

G. Certificates of Environmental Friendly
The Contractor shall submit to the Architect for Approval that the gypsum block used has been approved by Green Council and awarded the Certificate of Green Label under the category of Building Products using Recycled Materials (GL-008-009). The gypsum block manufacturer shall be certified by Green Council or relevant authorities on its operation system in procurement, production, delivery and disposal of wastage in full compliance with the requirements on environmental protection and sustainable development.

H. Project Reference and past performance

The AUGREEN block manufacturer shall have 10 years' experience in Hong Kong Market. A full list of past project reference, (including project address, name of owner / architect, contact of owner's representative, project completion year, etc) shall be submitted to Architect for Approval. Site visit and / or phone check may be conducted by the Architect to inspect and ascertain the past performance of the AUGREEN block were used in other projects. Should the Contractor failed to provide relevant information, the Architect shall reject the submitted material. Project reference from government bodies shall be an advantage.

I. Awards on Environmental Contribution
Award(s) from other institutes to demonstrate the manufacturer's contribution on environmental protection is preferable, including but not limited to ProductWise scheme, Hong Kong Award for Environmental Excellence, etc.

J. Project Reference and Past Performance
A full list of past project reference, (including project address, name of owner / architect, contact of owner's representative, project completion year, etc) shall be submitted to Architect for Approval. Site visit and / or phone check may be conducted by the Architect to inspect and ascertain the past performance of the gypsum block used in other projects. Should the Contractor failed to provide relevant information, the Architect shall reject the submitted material. Project reference from government bodies shall be an advantage.

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

(NOT USED)

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.TESTING

(NOT USED)

1.11 WARRANTY AND MAINTENANCE BOND

This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.

1.12 SPARE PART

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCTS

2.01 APPROVED MANUFACTURERS/SUPPLIERS

A. Subject to compliance with the Project Contract Document requirements, provide waterproofing, tanking lining, roofing system and required accessories from one or more of the following or equivalent approved by the Architect:

1. CaSO (HK) Engineering Company Limited
   Unit 1701-02, 17/F, 8 Observatory Road,
   Tsim Sha Tsui, Kowloon, Hong Kong.
   Tel: (852) 2887 8850 / 3188 4341
   Fax: (852) 3579 0447
   Email: joycechan@caso.com.hk

B. or approved equivalent

2.02 MATERIALS

A. The raw material of the AUGREEN gypsum block shall contain at least 50% in weight of recycled materials, including but not limited to chemical gypsum generated from Flue Gas Desulphurization (FGD) Plant of Electrical Power Plant. **No phosphgypsum** is allowed to use as the raw material of the blocks. The phosphorus content of the block should not exceed 0.05%.

B. The AUGREEN block shall be with tongue or groove profile on four sides, smooth finished surface and ready to receive the surface finish, fire rated to specified rating in the size of :

1. 75mm/80mm/100mm (T) x 500mm(H) x 666mm/ (W)
2. 150mm (T) x 500mm (H) x 333mm (W).Color of AUGREEN Block

Standard:	Gray White
Water Repelling:	Light Blue
High Density:	Yellowish

C. Supply of material shall be a consistent, even color range, batch to batch and within the batches.

D. Density of gypsum block wall shall be within the range of 900kg/m3 with a compressive strength not less than 5MPa. For High Density blocks, the density shall be 1,200kg/m^3 with the compressive strength not less than 14MPa.

E. The compressive strength of bonding agent shall not less than 19.6MPa, whereas the flexural strength shall not less than 9.2MPa.

F. The gypsum blocks shall be manufactured to the tolerance within +/- 2mm, +/- 1mm and +/- 0.5mm respectively for length, height and thickness.

G. The gypsum blocks shall be tested in accordance with BS 476 Part 22.

1. 2-hours FRP with conduits and socket boxes for 75mm/80mm thick
2. 4-hours FRP for 100mm/150mm thick

The FRP requirements shall refer to Architect's drawings. The Contractor shall submit certificate, test report and assessment reports regarding the fire rating performance to the Architect for approval. All tests shall be carried out by HOKLAS accredited Laboratories. Any cutting, drilling, and fixing of the gypsum block wall shall be carried out in strict accordance with the relevant fire testing reports, assessment reports, manufacturer's instruction and specification.

H. The gypsum block wall system shall be designed to withstand a lateral loading of 1.5kpa.

I. The proprietary based adhesive shall be made and supplied by the same gypsum block manufacturer to ensure their compatibility and overall performance of the gypsum block wall system. It shall be applied as a bonding agent for all interlocking joints of the gypsum block wall.

J. The raw material shall contain below contents not exceeding the below limits:

a.	lead (Pb)	<10
b.	Mercury (Hg)	<10
c.	Cadium (Cd)	<10
d.	Hexavalent chromium content (Cr(VI))	<10
e.	Arsenic content (As)	<10
f.	Selenium content (Se)	<10
g.	Volatile organic components (VOCs)	
h.	Formaldehyde (HCHO)	
i.	Radon (Rn)TIES AND ANCHORS	

PART 3 EXECUTION

3.01 PREPARATION

Before the installation of block wall, the conditions of the substrate in which the Work to be installed or affixed to shall be acceptable by the manufacturer. Nothing in the material, workmanship or construction method used in the Work of this section shall invalidate any manufacture’s warranties or reduce their warranty period.

3.02 INSTALLATION, GENERAL

A. Unless otherwise specified in drawings, for wet area installation, Water Repelling (WR) block shall be used for the lowest course of the block wall.

B. Water-Repelling (WR) block with minimum height of 500mm shall be used for the lowest course of the block wall in dry area if no concrete curb/kicker will be built.

C. For potential wet areas such as toilet/bathroom and kitchen, Water-Repelling (WR) block shall be used with the height higher than the required waterproofing system.

D. The contractor shall not use gypsum blocks that are cracked or damaged.

E. The gypsum block wall shall be erected in full height to the ceiling soffit. Opening for equipment and services installation shall be formed as marked on the completed wall.

F. For cutting of the gypsum blocks, the Contractor shall use power saws and ensure the edges in sharp and clean condition. All cutting edges of the block shall be in continuous pattern in order to match and fit the edge of adjoining construction. During the cutting process, the Contractor shall provide vacuum cleaner to minimize dust generated on site. The Contractor shall use full-size gypsum blocks without cutting whenever possible.

G. The gypsum block wall system shall be designed for an erection of wall up to a maximum height of 5.m (for 75mm), 5.5m (for 80mm Thick), 7.0m (for 100mm Thick), 8m (for 150mm Thick) without the need for installation of steel stiffeners. ICE calculation following DIN standard shall be submitted to Architect for approval.

H. Trenches for conduits /pipe works can be chased 3-5 days after the gypsum block wall construction. The contractor should fully cover the recess of conduits / pipe works with bonding agent after the installation of the E&M service.

I. No lintels are required to use within 1.5m (W) for AUGREEN Blocks in 75mm/80mm thick and 2m (W) for AUGREEN Blocks in 100mm/150mm thick.

J. The gypsum block wall system shall achieve a loading capacity of 1.4kN on pull out and 2.36kN on shear force. Submission of test report and/ or endorsement document from proprietary anchor suppliers (e.g. Hilti, Fisher, etc.) shall be submitted for approval.

3.03 TOLERANCES

Materials, products and supporting system specified in this section shall be installed to meet the following requirements on tolerances. All tolerances shall be non-cumulative.

A. Plumb
Variation from vertical lines and surfaces of column, internal corners and arises.
+/- 6mm for 3m heights
+/- 10mm for 6m heights
+/- 12mm for 12m heights or more

B. Level
Variation from level for bed joints and lines of exposed sills, parapets, horizontal grooves and other conspicuous lines
+/- 6mm for 6m lengths
+/- 12mm for 12m lengths or more

C. Notwithstanding the above, the Work shall have a satisfactory visual appearance, being square, regular, true to line, level and plane with a close fit to all junctions, all to match approved trial panel. Stepping at joints between AUGREEN Blocks or other sudden irregularity is not permissible.

3.04 CLEANING

(NOT USED)

3.05 PROTECTION

(NOT USED)

END OF SECTION